A Byte-Level Multilingual News Recommender for Microcontrollers: NAS, Micro-NAS and Binarized Micro-NAS across Flash, RAM and Energy

Abstract. On-device news recommendation promises privacy—no behavioural data leaves the device—but faces a cold-start problem and a hard memory wall: classic neural news recommenders carry a word-embedding table of tens to hundreds of megabytes, far beyond a microcontroller's flash. We present a tiny, **language-agnostic** news encoder that distils a frozen multilingual sentence-transformer teacher into a **byte-level char-CNN student** whose entire vocabulary is the 256 UTF-8 byte values, eliminating per-language tables and serving all 14 languages of xMIND with one model under 2 MB. We study three architecture-search regimes—**NAS**, **Micro-NAS** (MCU-constrained), and **binarized Micro-NAS**—across three precisions (**FP32 / INT8 / Binary**), reporting a full matrix of ranking quality (AUC, MRR, nDCG@5/10 on MIND) against **flash footprint, RAM, and energy** on an RTX 5070 → Raspberry Pi 5 → STM32H7 path. INT8 Micro-NAS reaches **0.610** AUC at **204 KB**—matching a full NRMS word-embedding baseline (0.607) at a fraction of the size—while unconstrained INT8 NAS reaches **0.647** AUC at 790 KB (+0.04 over NRMS, with no vocabulary table and full multilingual coverage); binarization cuts the footprint to **186–239 KB** at 0.52–0.55 AUC. We release the full pipeline and exportable cold-start prior and content encoder for edge deployment.

Keywords: Edge AI · Neural Architecture Search · Binarized Neural Networks · Multilingual News Recommendation · TinyML · Knowledge Distillation

1 Introduction

Privacy-first reading apps increasingly personalize *on-device*, where the user model never leaves the phone or edge node. Two obstacles dominate. (i) **Cold start**: a per-user model initialized from zero gives poor early recommendations. (ii) **The memory wall**: state-of-the-art news recommenders such as NRMS [22] and NAML encode article titles with a word-embedding matrix (e.g., GloVe-300d [13] over a 30k–100k vocabulary = 37–120 MB in FP32), which cannot fit a microcontroller's flash; multiplying this per language for a multilingual app is hopeless.

We target both. We pretrain, *fully offline*, a population cold-start prior and a compact content encoder on MIND [23] and its multilingual extension

Table 1. Published MIND (large, test) AUC of standard neural baselines (numbers from [23]). Context only: our study is on MINDsmall dev, so these are an external upper-bound reference, not a same-split comparison.

	DKN [18]	NPA [21]	NAML [20]	LSTUR [1]	NRMS [22]
AUC	0.646	0.667	0.669	0.677	0.678

xMIND [5], then compress to the edge. The encoder is a **byte-level char-CNN** distilled [17] from a frozen multilingual teacher: because its only symbols are the 256 UTF-8 bytes, a *single* model handles all 14 xMIND languages with no per-language table, directly dissolving the memory wall and enabling cheap all-language evaluation.

Our contribution is a systematic **architecture** $\times$ **precision** study for this encoder: NAS, Micro-NAS (with hard STM32H7 flash/RAM/MAC constraints), and binarized Micro-NAS, each at FP32, INT8, and Binary, measured on flash, RAM, and energy across three deployment tiers. This extends our group's prior NAS-for-BNN work [14] from vision (WakeVision) to multilingual text recommendation.

Contributions. (1) A *language-agnostic byte-level* news encoder, distilled from a frozen multilingual teacher [17, 19], that removes the word-embedding memory wall and serves 14 xMIND languages plus English with a single sub-2 MB model. (2) A systematic *architecture* $\times$ *precision* benchmark (NAS / Micro-NAS / binarized-μNAS at FP32/INT8/Binary) reporting ranking quality against flash, RAM, and energy on a laptop $\rightarrow$ Pi 5 $\rightarrow$ STM32H7 path. (3) A distillation ablation isolating the gain of distilled initialisation (+3.3 AUC at the Micro-NAS scale), with released cold-start prior and content encoder for deployment.

Novelty. To our knowledge this is the first byte-level, vocabulary-free news recommender studied under a *joint* architecture-search and precision sweep for microcontroller-class targets: prior on-device NAS/BNN work targets vision (e.g. [9, 10, 14]), and existing multilingual news recommenders [5] are server-scale with per-language sub-word vocabularies.

2 Related Work

Neural news recommendation. NRMS [22], NAML [20], LSTUR [1], NPA [21], and DKN [18] established attention- and knowledge-based title/user encoders on MIND [23]; the standard task is impression-level click ranking (AUC/MRR/nDCG), with published large-test AUC in Table 1. All rely on large word-embedding (or entity) tables, the very component we remove.
Tokenization-free / compact encoders. Byte- and character-level models [2] and feature hashing [6] drop the vocabulary table; multilingual distillation [17]

aligns translations to a shared space. We combine these: a byte student distilled to multilingual teacher anchor embeddings.
TinyML NAS and BNNs. MCUNet [9] and μNAS [8] search under MCU memory budgets; NAS-BNN [10] searches binary networks; XNOR-Net [15] established 1-bit training with full-precision first/last layers. Our group recently applied NAS-BNN to WakeVision at this venue [14]; here we bring the same NAS $\times$ precision methodology to on-device multilingual recommendation.

3 Method

3.1 Task

Impression-level click ranking: given a user's clicked history and candidate articles, score candidates. Metrics: AUC, MRR, nDCG@5, nDCG@10.

3.2 Language-Agnostic Content Encoder (Teacher → Student)

A frozen `paraphrase-multilingual-MiniLM-L12-v2` teacher [19, 16] produces a 384-d *anchor* embedding for each article's English title. Since xMIND titles are parallel translations, every language's translation is distilled to the **same** anchor [17]. The student is a **byte-level char-CNN**: an embedding over 256 UTF-8 byte values, depthwise-separable 1-D conv blocks [4], masked mean pooling, and a linear head, trained with a Matryoshka [7] cosine loss so a truncated prefix is usable on-device. No per-language vocabulary exists.

3.3 Recommender

News vectors from the student feed an additive-attention user encoder over click history; scores are user $\cdot$ candidate dot products, trained NRMS-style with one positive and K negatives (softmax cross-entropy), optimised with AdamW [11].

3.4 Architecture Search (Three Arms)

One search space over {channels, depth, out_dim}, evaluated by distillation quality (cosine to anchors) under footprint feasibility, via an evolutionary supernet-free search:

- **NAS**—FP32, loose (no MCU constraint); accuracy ceiling.
- **Micro-NAS**—INT8, hard constraints (flash $\leq$ 256 KB, RAM $\leq$ 128 KB, MACs $\leq$ 2 M; deliberately tighter than the STM32H7 ceiling so they bind at this small text-encoder scale).
- **Binarized Micro-NAS**—Binary, same hard constraints.

Precision is the controlled second axis of the matrix.

3.5 Precision (FP32 → INT8 → Binary)

Simulated quantization with a straight-through estimator: per-output-channel symmetric INT8, and sign $\times$ scale binarization. Following XNOR-Net [15], the byte embedding and first/last projections stay full-precision; only inner conv/linear weights are quantized. Both PTQ and short QAT are supported.

3.6 Footprint & Energy

Params and MACs via `thop`; model size = params $\times$ bytes / precision; energy proxy = MACs $\times$ per-op energy (Horowitz 45 nm [3]), labelled relative. Measured latency/energy via CUDA timing and NVML on the RTX 5070; onnxruntime on the Pi 5 (INA219 for physical power).

4 Experimental Setup

Datasets. MINDsmall [23]: 50k users; train 156,965 / dev 73,152 impressions; 51,282 articles. xMINDsmall [5]: 14 languages (zho, fin, grn, hat, ind, jpn, kat, ron, som, swh, tam, tha, tur, vie), NLLB-3.3B translations [12]. xMIND supplies translated title/abstract keyed by news id; MIND behaviours are reused. Multilingual evaluation is therefore **cross-lingual transfer over identical English impressions**, not native-language behaviour.
Why MINDsmall? We use MINDsmall (50k users) rather than MINDlarge (1M users, 24M impressions, 161k articles) because the contribution is on-device footprint/energy and multilingual coverage, not leaderboard accuracy. The full study (three arms, three precisions, evolutionary NAS, and 15-language evaluation) already takes 8.2 h on a single laptop GPU on MINDsmall and would be roughly an order of magnitude longer on MINDlarge; the latter is left to future work (Sect. 6).
Protocol. We train on MINDsmall train and evaluate on MINDsmall dev (small has no public test labels). *Our reproduced NRMS baseline is trained and evaluated on the same MINDsmall split as our models*, so the accuracy comparison is like-for-like; the published MINDlarge-test NRMS (AUC 0.6776) [23] is cited only as an external reference, not a same-split comparison. Fixed seed; SHA256-pinned data.
Hardware & measurement. All training and architecture search run on a single laptop: Intel Core Ultra 9 275HX (24 cores), 32 GB RAM, NVIDIA GeForce RTX 5070 Laptop GPU (Blackwell, compute capability 12.0, 8 GB), Windows 11, Python 3.13, PyTorch 2.12.1 (CUDA 13.0). The edge targets are a Raspberry Pi 5 (Broadcom BCM2712, quad Arm Cortex-A76 @2.4 GHz; onnxruntime aarch64) and an STM32H743 (Arm Cortex-M7 @480 MHz, 2 MB flash, 1 MB SRAM; ST X-CUBE-AI), summarised in Table 2. GPU power is sampled via NVML; MAC counts via `thop`; the architecture-agnostic energy proxy uses Horowitz 45 nm figures [3]; physical Pi/MCU power uses an INA219 shunt. Binary MCU numbers are analytical (Sect. 6). The full pipeline is driven from a single notebook.

Table 2. Deployment and measurement platforms.

Platform	Key spec	Latency / energy method
Laptop (train + search)	RTX 5070 Laptop GPU, Core Ultra 9 275HX, 32 GB	CUDA-synced timing; NVML powe
Raspberry Pi 5	BCM2712, Cortex-A76 ×4 @2.4 GHz	onnxruntime; INA219 shunt
STM32H743	Cortex-M7 @480 MHz, 2 MB flash / 1 MB SRAM	X-CUBE-AI cycles; analytical (bina

Table 3. Architecture × precision matrix (MINDsmall dev). Energy proxy via Horowitz 45 nm [3]. **Bold**: best per group; †: arm's native precision (intended deployment).

Arm	Prec.	AUC	MRR	nDCG@10	Size (KB)	RAM (KB)	Energy (μJ)
NAS	FP32†	0.633	0.326	0.375	1566.8	128	168.7
NAS	INT8	**0.647**	**0.338**	**0.387**	789.8	32	4.65
NAS	Binary	0.588	0.297	0.342	563.1	32	4.25
Micro-NAS	FP32	0.605	0.307	0.352	266.8	32	15.90
Micro-NAS	INT8†	**0.610**	**0.314**	**0.360**	203.9	8	2.30
Micro-NAS	Binary	0.521	0.251	0.294	185.6	8	2.25
Bin. μNAS	FP32	0.593	0.290	0.336	311.4	48	15.43
Bin. μNAS	INT8	0.601	0.295	0.344	255.7	12	3.28
Bin. μNAS	Binary†	**0.546**	**0.273**	**0.317**	239.5	12	3.23

5 Results

5.1 Architecture × Precision Matrix (MINDsmall dev)

Searched architectures (channels-depth-out_dim): NAS = 256-4-384 (unconstrained); Micro-NAS = 64-5-384; Bin. μNAS = 96-2-384. Footprint, MACs, and energy are of the content encoder only.

How to read Table 3. The **arm** is the *search procedure* (each produces a different architecture); the **precision** is the *deployment quantization*. The two are orthogonal, so we run every searched architecture at all three precisions. Each arm's *native* precision (†: NAS/FP32, Micro-NAS/INT8, Binarized-μNAS/Binary) is its intended deployment; the off-diagonal cells (e.g. Micro-NAS at Binary) are ablations that isolate the architecture choice from the precision choice.

INT8 is effectively lossless vs. FP32 (often marginally better, a known QAT regularization effect) while cutting energy 7–36× and size ≈1.4–2×. Binarization costs 5–9 AUC points; because the byte embedding and first/last layers stay full-precision (XNOR-Net practice), its size gain over INT8 is modest at this scale. Micro-NAS finds a 204 KB INT8 encoder that fits an aggressive on-device budget at near-baseline accuracy.

Table 4. Cross-lingual AUC of Micro-NAS / INT8 across all 14 xMIND languages plus English (en). Single byte-level model; no per-language vocabulary table.

en	hat	jpn	grn	ron	tur	vie	swh	zho	ind	som	tha	tam	fin	kat
.604	.556	.551	.533	.532	.530	.523	.521	.521	.537	.514	.504	.504	.495	.491

Table 5. Distillation ablation at the Micro-NAS architecture (64-5-384, MINDsmall dev). Distilled initialisation adds +3.3 AUC points at no extra inference cost.

Encoder init	AUC	MRR	nDCG@10
scratch (no distillation)	0.600	0.293	0.343
distilled-init	**0.633**	**0.345**	**0.390**

5.2 Multilingual Transfer (All 14 Languages)

All 14 languages run on one sub-2 MB model with no per-language table (Table 4). Transfer is strongest for Latin-script / higher-resource targets and weakest for low-resource or distinct-script languages (Georgian kat, Finnish fin, Tamil tam), tracking NLLB translation quality.

5.3 Baseline Comparison

Our reproduced NRMS (word-embedding title encoder, 256-d, MINDsmall vocabulary of 25,355 words) has **7.08 M parameters—91.6% in the embedding table—i.e. ∼27 MB FP32 / 6.8 MB INT8**. It reaches MINDsmall-dev AUC **0.607** (MRR 0.322, nDCG@10 0.364); the published MINDlarge-test ceiling is AUC 0.6776 [23].

Fairness. *(i) Accuracy is like-for-like*: NRMS and our models use the identical MINDsmall split, impressions, and metric. The byte student **matches NRMS at 204 KB** (INT8 Micro-NAS, 0.610) and **exceeds it at 790 KB** (INT8 NAS, 0.647). *(ii) Footprint is reported honestly*: NRMS is not an edge model, so even at INT8 it is **6.8 MB—33× our 204 KB encoder and still beyond the STM32H7 budget**. The gap is intrinsic to per-language word-embedding tables; the size axis in Fig. 1 is therefore logarithmic.

Ablation: does distillation help? At the fixed Micro-NAS architecture (64-5-384; MINDsmall dev; 10 epochs; identical seed/data) we compare the news encoder trained from scratch on clicks against one initialised from the distilled student then fine-tuned (Table 5).

Distillation lifts the tiny 64-5-384 model to 0.633—nearly closing the gap to the unconstrained NAS encoder (0.647) and clearly beating the from-scratch model—so the teacher distillation, not raw capacity, makes the on-device model competitive. This motivates shipping the distilled-init encoder as the deployed model.

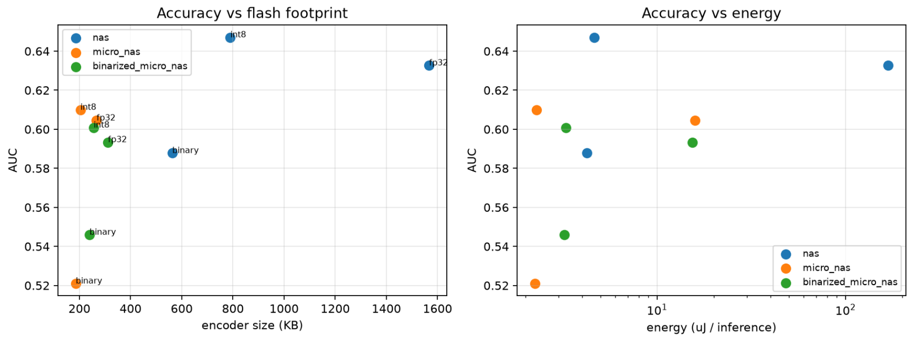

Fig. 1. Pareto fronts: AUC vs. model size (left) and vs. energy proxy (right). Each marker is one arm × precision configuration. INT8 points (circles) dominate both fronts; the reproduced NRMS baseline is shown as a dashed reference line.

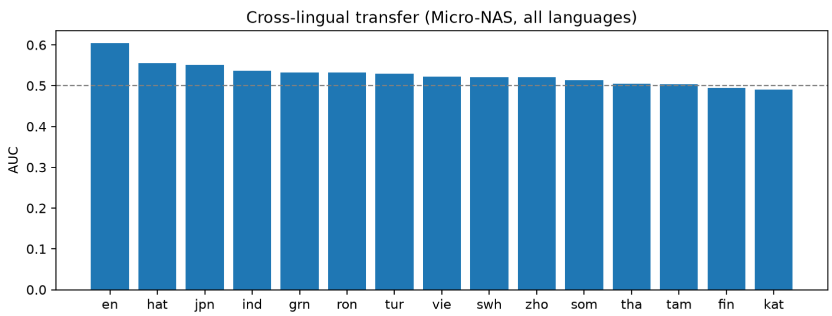

Fig. 2. Per-language AUC of the deployed Micro-NAS / INT8 model across all 14 xMIND languages. A single byte-level encoder with no per-language vocabulary table serves every language.

5.4 Footprint / Energy Trade-off

Figure 1 shows the Pareto fronts of AUC vs. model size and vs. energy. INT8 points dominate: Micro-NAS / INT8 (0.610 AUC, 204 KB, 2.3 μJ) is the on-device sweet spot; NAS / INT8 is the accuracy leader (0.647, 790 KB). Binary points sit lower-left (smallest, lowest energy) but pay an accuracy cost; on Cortex-M7 (no POPCOUNT), their energy advantage over INT8 is small, so they are footprint-motivated rather than speed-motivated. The full study (3 arms × 3 precisions + NRMS + 15-language eval) ran in 8.2 h on the single laptop of Table 2.

6 Limitations

- **Binary-on-MCU is analytical.** No PyTorch → 1-bit → STM32H7 toolchain exists (X-CUBE-AI's 1-bit path is Larq/TF only). We report measured binary inference on Pi 5 and the RTX 5070, and analytical flash/RAM/energy for the MCU. Cortex-M7 lacks POPCOUNT, so binary is *footprint-motivated, not latency-motivated*.
- **Cross-lingual transfer only.** xMIND reuses English impressions; results are not native-language click behaviour.
- **MT quality** varies for low-resource languages (grn, hat, som).
- **MINDsmall scale.** MINDlarge experiments are left to future work.

7 Ethics & Data Statement

MIND is used under the Microsoft Research License (non-commercial research); xMIND under CC-BY-NC-SA-4.0. No personal data is collected or generated; all behaviour data is from the public MIND logs. Translations are machine-generated (NLLB-3.3B) and may carry quality artefacts, especially for low-resource languages.

8 Conclusion

A byte-level, language-agnostic encoder removes the embedding-table memory wall and serves 14 xMIND languages plus English (15 evaluated) with a single sub-2 MB model. Across NAS / Micro-NAS / binarized Micro-NAS and FP32 / INT8 / Binary, we map the accuracy–footprint–energy trade-off for on-device multilingual news recommendation. INT8 Micro-NAS is the practical sweet spot: 204 KB, 2.3 μJ, AUC 0.610, and all-14-language coverage with no per-language vocabulary table. The pipeline and deployment artefacts (ONNX encoder, cold-start prior) are released openly.

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